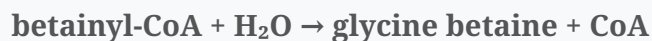


PP_0301 (Q88R33) in *Pseudomonas putida* KT2440: A Cytoplasmic Betainyl-CoA Thioesterase of the L-Carnitine Catabolic Pathway

Summary

The gene **PP_0301** (UniProt **Q88R33**; OrderedLocusNames PP_0301) of *Pseudomonas putida* KT2440 encodes a small (156-residue), single-domain, **HotDog-fold acyl-CoA thioesterase** belonging to the 4-hydroxybenzoyl-CoA thioesterase (4HBT) family. Its physiological function is that of a **betainyl-CoA thioesterase** (gene name **cdhB**; EC 3.1.2.33; KEGG ortholog **K27492**), catalyzing the hydrolysis of the CoA thioester of glycine betaine:



This reaction is the third, CoA-releasing step of the **L-carnitine catabolic pathway** encoded by the compact, co-oriented operon **PP_0301–PP_0305**. Within this pathway, L-carnitine is first oxidized to 3-dehydrocarnitine by carnitine 3-dehydrogenase (lcdH/cdhA, PP_0302, EC 1.1.1.108), the 3-dehydrocarnitine is then cleaved by the β -keto-acid cleavage enzyme cdhC (PP_0303, EC 2.3.1.317) to yield **betainyl-CoA** (with concomitant acetoacetate formation from acetyl-CoA), and finally PP_0301/cdhB hydrolyzes betainyl-CoA to release **glycine betaine** and free CoA. Glycine betaine is a central branch-point metabolite that is subsequently funneled into the conserved, cytoplasmic glycine-betaine demethylation pathway (GB $\rightarrow$ dimethylglycine $\rightarrow$ sarcosine $\rightarrow$ glycine), enabling the bacterium to use L-carnitine as a source of carbon and nitrogen.

A key outcome of this investigation is an **annotation correction**. The legacy name carried in the genome submission — "betainyl-CoA thiolase" with EC 2.3.1.- (an acyltransferase/thiolase class) — is a **misannotation**. Every domain and family assignment for Q88R33 is unanimous for a HotDog-fold thioester **hydrolase** (EC class 3.1.2.-), and the protein's small size (~156 aa, a single HotDog domain) is inconsistent with genuine thiolases (~400 aa, two-domain

acyltransferases). The correct designation is a **betainyl-CoA thioesterase (hydrolase)**. This functional assignment is strongly supported by 1:1 orthology to the experimentally validated *Pseudomonas aeruginosa* gene PA5385 (cdhB), which resides in a transposon-screen-confirmed carnitine-catabolism locus.

The protein is a soluble, **cytoplasmic** enzyme — it has no signal peptide or transmembrane segments (confirmed by a hydropathy scan), and its CoA-thioester substrate is strictly intracellular. Its L-carnitine substrate enters the cell through the co-encoded ABC transporter (substrate-binding protein PP_0304). An experimental crystal structure of the *P. putida* protein itself exists (PDB 2HLJ, 2.00 Å), confirming the HotDog thioesterase fold.

Key Findings

F001 — PP_0301 is a HotDog-fold thioesterase, not a thiolase (annotation corrected)

UniProt Q88R33 is a small **156-amino-acid single-domain protein**. The domain and family assignments across all major resources converge unanimously on a **HotDog-fold acyl-CoA thioesterase**:

Resource	Assignment
InterPro	IPR050563 (4-hydroxybenzoyl-CoA thioesterase family) + IPR029069 (HotDog domain superfamily)
Pfam	PF13279 (4HBT_2)
CDD	cd00586 (4HBT)
Gene3D	3.10.129.10 (HotDog Thioesterase)
SUPFAM	SSF54637 (Thioesterase/thiol ester dehydrase-isomerase)
PANTHER	PTHR31793 (4-hydroxybenzoyl-CoA thioesterase family)
eggNOG	COG0824 (predicted thioesterase)
GO (molecular function)	GO:0047617 (fatty acyl-CoA hydrolase activity)

Critically, an **experimental crystal structure exists: PDB 2HLJ**, an X-ray structure at 2.00 Å resolution (chains A = residues 1–156), solved by the Joint Center for Structural Genomics (JCSG/PSI-2) and titled "*Crystal structure of a putative thioesterase from Pseudomonas putida KT2440.*" This directly confirms the HotDog thioesterase fold experimentally for the exact protein in question.

The legacy genome-submission name "betainyl-CoA **thiolase**" (EC 2.3.1.-) is therefore a **misnomer**. Thiolases and other EC 2.3.1.- acyltransferases *transfer* an acyl group and are typically ~400-residue, two-domain enzymes. PP_0301 is instead a **hydrolase** (EC 3.1.2.-): it cleaves a thioester bond with water, releasing a free acid and CoA. Its small single-domain architecture is diagnostic of the HotDog thioesterase superfamily rather than the thiolase superfamily.

The mechanistic identity of the family is well established in the literature. HotDog-fold thioesterases catalyze hydrolysis of thioester bonds of acyl-CoA substrates (PMID: 26067557: "*It belongs to the hotdog-fold thioesterase superfamily and catalyzes hydrolysis of thioester bonds of acyl-CoA in vitro, while its in vivo function remains unrevealed.*"). Acyl-CoA thioesterases more broadly "*catalyze the hydrolysis of fatty acyl-CoA to free fatty acid and CoA and thereby regulate lipid metabolism and cellular signaling*" (PMID: 17563367), and the structural determinants for substrate recognition and catalysis by these enzymes have been dissected in the *E. coli* paralogs YbdB and YdiI (PMID: 25010423: "*the structural determinants for substrate recognition and catalysis in two hotdog-fold thioesterase paralogs, YbdB and YdiI from Escherichia coli, are identified and analyzed*").

F002 — PP_0301 = cdhB, betainyl-CoA thioesterase (EC 3.1.2.33)

KEGG assigns **ppu:PP_0301** to ortholog **K27492 = cdhB = "betainyl-CoA thioesterase [EC:3.1.2.33]."** The EC 3.1.2.33 entry (reaction RN R13307) defines the catalyzed reaction as:

betainyl-CoA + H₂O = glycine betaine + CoA (systematic name: *betaine-CoA hydrolase*)

The KEGG comment notes that "*the enzyme, characterized from the bacterium Pseudomonas aeruginosa, is involved in an L-carnitine degradation pathway.*" The **substrate specificity** is the CoA thioester of glycine betaine (betainyl-CoA); the enzyme hydrolyzes this thioester to release free glycine betaine plus CoA. This assignment is fully consistent with (i) the HotDog 4HBT fold — a small, single-domain thioesterase scaffold — and (ii) the substrate implied by the protein's

own legacy name ("betainyl-CoA"). The reaction it catalyzes matches the general acyl-CoA thioesterase chemistry of the family (hydrolysis of an acyl-CoA to free acid + CoA), specialized here for the quaternary-ammonium acyl group of glycine betaine.

F003 — PP_0301 lies in a carnitine-catabolism operon (cdhABC / PP_0301–PP_0305)

Genomic context is highly informative. KEGG genomic positions place PP_0301 (complement 362036–362506) within a tight, co-oriented gene cluster:

Locus	Gene	KEGG ortholog	Enzyme / role	EC
PP_0301	cdhB	K27492	Betainyl-CoA thioesterase	3.1.2.33
PP_0302	lcdH / cdhA	K17735	Carnitine 3-dehydrogenase	1.1.1.108
PP_0303	cdhC	K27837	3-dehydrocarnitine:acetyl-CoA trimethylamine transferase (BKACE)	2.3.1.317
PP_0304	—	K02002	Glycine betaine/carnitine ABC-transporter substrate-binding protein	—
PP_0305	cdhR	K17736	AraC-family carnitine-catabolism transcriptional activator	—
PP_0298	—	K21826	AraC-family glycine-betaine-responsive activator	—

STRING (160488.PP_0301) corroborates the functional linkage: its top-scoring partner is **lcdH** (score 0.995), followed by **caiX** (carnitine transport, 0.632) and **cdhR** (0.592). Integrating these data yields the reconstructed pathway:

```

L-carnitine
  | cdhA / lcdH (PP_0302, EC 1.1.1.108; NAD+ → NADH)
  ▼
3-dehydrocarnitine
  | cdhC (PP_0303, EC 2.3.1.317; + acetyl-CoA → acetoacetate)
  ▼
betainyl-CoA
  | cdhB / PP_0301 (EC 3.1.2.33; + H2O)
  ▼
glycine betaine + CoA

```

PP_0301 catalyzes the **final, CoA-releasing step** of this three-enzyme route, regenerating free CoA and liberating glycine betaine for downstream catabolism.

F004 — Cytoplasmic localization; product feeds the GbdR-regulated demethylation pathway

PP_0301 is a **soluble, single-domain cytoplasmic enzyme**. It carries no signal peptide and no transmembrane segments (confirmed by a hydropathy scan), and its CoA-thioester substrate is strictly intracellular (CoA thioesters do not cross membranes). The upstream substrate, L-carnitine, is imported via the co-encoded ABC transporter (substrate-binding protein PP_0304). Thus the enzyme performs its chemistry in the **cytoplasm**.

Its product, **glycine betaine (GB)**, is the branch-point metabolite of quaternary-amine catabolism in pseudomonads. In the closely related *P. aeruginosa*, GB is degraded in the cytoplasm by a multistep demethylation route (GB → dimethylglycine → sarcosine → glycine), which lets the bacterium use GB — and its precursors choline and carnitine — as sole sources of carbon and nitrogen (PMID: 23354714: "*a variety of soil- and water-dwelling bacteria have catabolic pathways for the multistep conversion of choline, via GB, to glycine and can thereby use choline and GB as sole sources of carbon and nitrogen*"). This catabolic locus is controlled by the AraC-family regulator **GbdR**, whose regulon "*includes the genes encoding GB, dimethylglycine, sarcosine, glycine, and serine catabolic enzymes and the BetX and CbcXWV quaternary amine transport proteins*" (PMID: 24097953). This mirrors the CdhR/AraC regulator (PP_0305) that governs the carnitine branch in *P. putida*.

F005 — Primary literature confirms the carnitine → 3-dehydrocarnitine → glycine betaine route

The two enzymatic steps flanking and including PP_0301 are anchored in primary experimental literature:

- **Wargo & Hogan 2009 (PMID: 19406895)** performed a transposon-mutant genetic screen in *P. aeruginosa* PA14 that experimentally established the carnitine catabolic route: "*carnitine is converted to 3-dehydrocarnitine (3-dhc) which is in turn metabolized to glycine betaine (GB), an intermediate metabolite in the catabolism of carnitine to glycine.*" They also "*confirmed that an intact GB catabolic pathway is required for growth on carnitine,*" showing that carnitine catabolism funnels through glycine betaine — the very product of PP_0301.
- **Bastard et al. 2014 (PMID: 24240508)** defined the **DUF849 / β -keto acid cleavage enzyme (BKACE)** family that provides the EC 2.3.1.317 activity (cdhC): "*we investigated the DUF849 Pfam family and unearthed 14 potential new enzymatic activities, leading to the designation of these proteins as β -keto acid cleavage enzymes.*" This is the enzyme that generates the **betainyl-CoA substrate** hydrolyzed by PP_0301.
- **Wargo, Szwergold & Hogan 2008 (PMID: 17951379)** biochemically defined the downstream glycine-betaine demethylation genes and the AraC regulator GbdR in *P. aeruginosa*: "*Glycine betaine (GB), which occurs freely in the environment and is an intermediate in the catabolism of choline and carnitine, can serve as a sole source of carbon or nitrogen in Pseudomonas aeruginosa.*"

The KT2440 cluster PP_0301–PP_0305 (cdhB, lcdH/cdhA, cdhC, transporter, cdhR) is the direct ortholog of the *P. aeruginosa* carnitine-catabolism locus.

F006 — PP_0301/cdhB is the direct ortholog of *P. aeruginosa* PA5385, within an experimentally validated locus

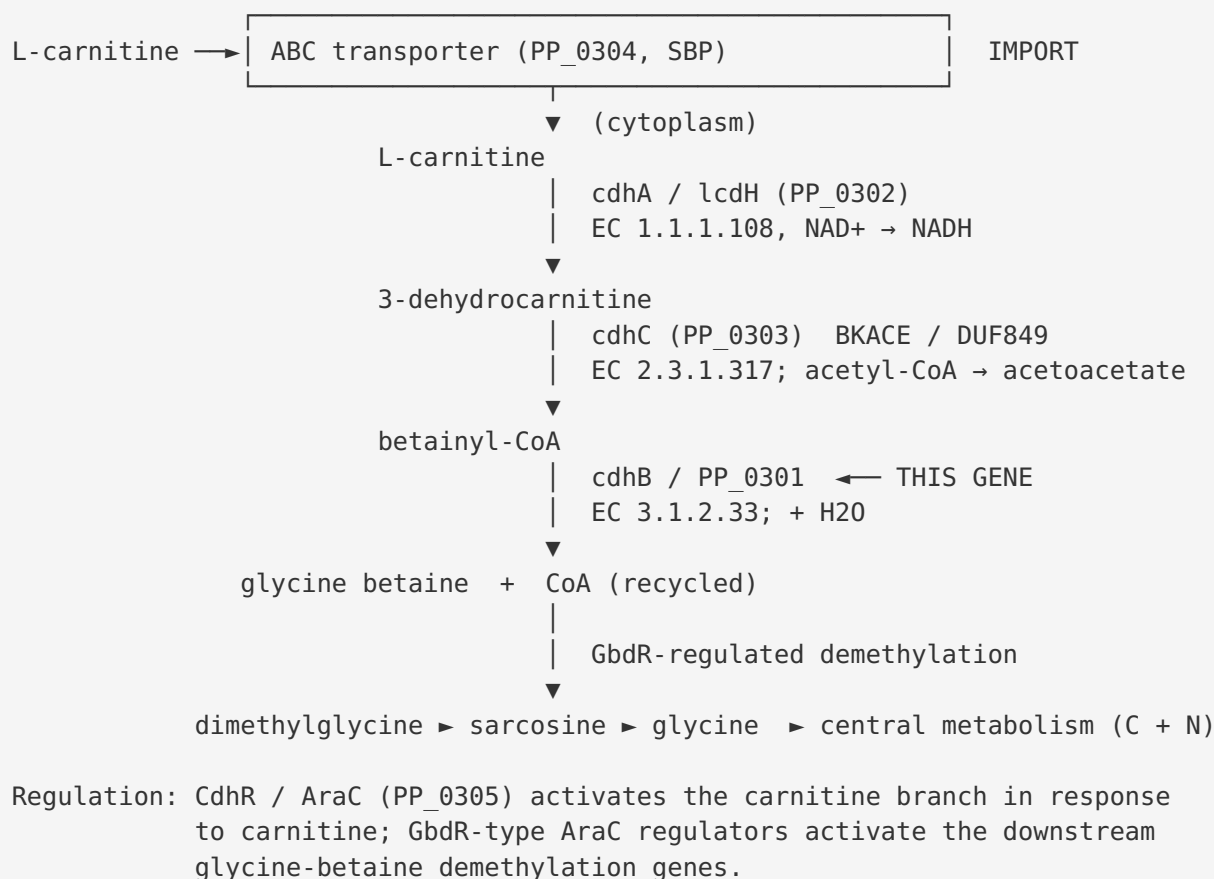
KEGG orthology links K27492 (cdhB) to *P. aeruginosa* PAO1 gene **PA5385** (and PA14 gene PA14_71120). The adjacent cluster resolves cleanly into an ortholog-by-ortholog match:

<i>P. putida</i> KT2440	<i>P. aeruginosa</i> PAO1	Gene	Function
PP_0301	PA5385	cdhB	Betainyl-CoA thioesterase (K27492, EC 3.1.2.33)
PP_0302	PA5386	lcdH/ cdhA	Carnitine 3-dehydrogenase (K17735, EC 1.1.1.108)
PP_0303	PA5387	cdhC	BKACE (K27837, EC 2.3.1.317)
PP_0305	PA5389	cdhR	AraC regulator (K17736)

This is precisely the **"PA5388–PA5384 region"** (with adjacent regulator PA5389) that Wargo & Hogan 2009 identified by transposon screen as containing the carnitine-dehydrogenase homologue and genes required for growth on carnitine ([PMID: 19406895](#): "*The PA5388-PA5384 region contains the predicted *P. aeruginosa* carnitine dehydrogenase homologue along with other genes required for growth on carnitine.*"), with PA5389 required to induce those transcripts in response to carnitine. Consequently, *P. putida* PP_0301 (cdhB) is **1:1 orthologous to an experimentally validated carnitine-catabolism gene (PA5385)**, and its operon PP_0301–PP_0305 mirrors PA5385–PA5389. This transfers strong experimental support from the *P. aeruginosa* system to the *P. putida* annotation.

Mechanistic Model / Interpretation

PP_0301 is best understood as the **CoA-recycling hydrolase of the carnitine catabolic operon** in *P. putida* KT2440. The pathway integrates transport, oxidation, thioester-generating cleavage, and thioester hydrolysis:



Why the "thiolase" name is wrong and the "thioesterase" assignment is right. The genome-submission label reflects an early, sequence-blind guess. Three independent lines of evidence overturn it:

- 1. Fold / size** — Q88R33 is a 156-aa single HotDog domain (confirmed experimentally by PDB 2HLJ), not a ~400-aa two-domain thiolase. Every domain database (InterPro, Pfam, CDD, Gene3D, SUPFAM, PANTHER, eggNOG) classifies it as a 4HBT HotDog thioesterase.
- 2. Chemistry** — HotDog 4HBT enzymes *hydrolyze* acyl-CoA thioesters (EC 3.1.2.-), releasing free acid + CoA; thiolases (EC 2.3.1.-) instead *transfer* acyl groups. The KEGG-assigned EC 3.1.2.33 reaction (betainyl-CoA + H₂O → glycine betaine + CoA) is a hydrolysis.
- 3. Pathway logic and orthology** — The enzyme sits immediately downstream of a BKACE that *produces* betainyl-CoA, and it is 1:1 orthologous to the experimentally validated *P. aeruginosa* cdhB (PA5385). A thioesterase that liberates glycine betaine + CoA is exactly what the pathway requires at this position; a thiolase would not fit.

Biological role. By hydrolyzing betainyl-CoA, PP_0301 (i) liberates glycine betaine for entry into the demethylation pathway, and (ii) regenerates free CoA, which is essential for continued flux through the acetyl-CoA-dependent cdhC step. The enzyme thereby couples carnitine breakdown to the universal quaternary-amine catabolic hub (glycine betaine → glycine), allowing *P. putida* to exploit carnitine as a carbon and nitrogen source in the soil and rhizosphere environments where carnitine and its relatives are common.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
19406895	<i>Identification of genes required for P. aeruginosa carnitine catabolism</i>	Primary experimental anchor. Transposon screen establishing carnitine → 3-dehydrocarnitine → glycine betaine; identifies the PA5388–PA5384 region (containing PP_0301's ortholog PA5385) as required for growth on carnitine.
24240508	<i>Revealing the hidden functional diversity of an enzyme family</i>	Defines the DUF849 / BKACE family that provides cdhC (EC 2.3.1.317), the enzyme generating PP_0301's betainyl-CoA substrate.
17951379	<i>Two gene clusters and a regulator for P. aeruginosa glycine betaine catabolism</i>	Confirms glycine betaine (PP_0301's product) is a carnitine-catabolism intermediate usable as sole C/N source; defines downstream demethylation genes and GbdR.
24097953	<i>Characterization of the GbdR regulon in P. aeruginosa</i>	Documents downstream GB→DMG→sarcosine→glycine catabolic enzymes and AraC-family (GbdR) regulation receiving PP_0301's product.
23354714	<i>Homeostasis and catabolism of choline and glycine betaine</i>	Establishes the cytoplasmic multistep demethylation of glycine betaine to glycine in soil/water pseudomonads.
26067557	<i>Crystal structure of zebrafish fTHEM2</i>	Establishes that HotDog-fold thioesterases (PP_0301's family) hydrolyze acyl-CoA thioester bonds.
17563367	<i>Structural basis for tandem hotdog domains in ACOT7</i>	Defines the HotDog acyl-CoA thioesterase reaction class (acyl-CoA → free acid + CoA).
25010423	<i>Structure and catalysis in E. coli YdiI and YbdB</i>	Confirms defined active-site residues drive thioester hydrolysis in the HotDog mechanistic class.

Additional context papers on choline/glycine-betaine physiology and sarcosine regulation in *P. aeruginosa* and *Burkholderia* (PMID: [22753069](#), PMID: [23457628](#), PMID: [26503852](#), PMID: [27381916](#)) reinforce the conservation and regulatory logic of the quaternary-amine catabolic network into which PP_0301 feeds.

Nature of the evidence. The strongest evidence for PP_0301's specific function is **inference by orthology** to an experimentally characterized system: the enzyme is a 1:1 ortholog of *P. aeruginosa* cdhB/PA5385, whose operon was validated genetically. The reaction assignment (EC 3.1.2.33) derives from biochemical characterization of the *P. aeruginosa* enzyme (per KEGG). Structural evidence is direct and experimental for the *P. putida* protein itself (PDB 2HLJ crystal structure). There is, to date, no published *direct in vitro* enzymatic assay of the *P. putida* KT2440 PP_0301 protein specifically.

Limitations and Knowledge Gaps

1. **No direct biochemical assay of PP_0301 itself.** The EC 3.1.2.33 activity was characterized in the *P. aeruginosa* enzyme; the *P. putida* ortholog's activity and kinetic parameters (kcat, Km for betainyl-CoA) are inferred, not measured. The PDB 2HLJ structure was deposited as a "putative thioesterase" without a bound substrate or functional assay.
 2. **Substrate specificity breadth is unquantified.** HotDog thioesterases can be promiscuous. Whether PP_0301 is strictly specific for betainyl-CoA or also hydrolyzes other short-chain or quaternary-amine acyl-CoAs is unknown.
 3. **Regulation in *P. putida* is inferred from *P. aeruginosa*.** The role of PP_0305 (CdhR) and PP_0298 (a glycine-betaine-responsive AraC regulator) in inducing PP_0301 in KT2440 is predicted by orthology and operon context, not directly demonstrated.
 4. **Phenotype not directly tested in KT2440.** A PP_0301 deletion has not been reported; its necessity for growth of *P. putida* on carnitine is inferred from the *P. aeruginosa* transposon screen.
 5. **Legacy annotation persistence.** Public databases still carry the "betainyl-CoA thiolase / EC 2.3.1.-" label; downstream tools relying on this may propagate the misannotation.
-

Proposed Follow-up Experiments / Actions

1. **In vitro enzyme assay.** Express and purify recombinant PP_0301 (Q88R33) and measure thioesterase activity against **betainyl-CoA** and a panel of acyl-CoA substrates (acetyl-CoA, short-chain acyl-CoAs, 4-hydroxybenzoyl-CoA) to confirm EC 3.1.2.33 and quantify substrate specificity (kcat/Km).

2. **Genetic knockout / complementation.** Construct a Δ PP_0301 mutant in *P. putida* KT2440 and test growth on L-carnitine and glycine betaine as sole C/N sources; complement to confirm the phenotype maps to PP_0301. Metabolite profiling should show betainyl-CoA accumulation in the mutant.
3. **Structure–function analysis.** Use PDB 2HLJ to identify the catalytic residue(s) (typically an Asp/Glu or backbone-water general base in 4HBT enzymes) and test by site-directed mutagenesis.
4. **Transcriptional regulation.** Test induction of the PP_0301–PP_0305 operon by carnitine and glycine betaine, and confirm CdhR (PP_0305) / PP_0298 dependence via reporter fusions and EMSAs.
5. **Database correction.** Submit a curation request to UniProt/KEGG to update Q88R33 from "betainyl-CoA thiolase (EC 2.3.1.-)" to "**betainyl-CoA thioesterase / cdhB (EC 3.1.2.33)**" with the supporting orthology and structural evidence.

Conclusion

PP_0301 (Q88R33) is a cytoplasmic, single-domain HotDog-fold (4HBT) acyl-CoA thioesterase functioning as a **betainyl-CoA thioesterase (cdhB; EC 3.1.2.33)**. It hydrolyzes betainyl-CoA to glycine betaine + CoA, performing the third, CoA-releasing step of the L-carnitine catabolic pathway encoded by the PP_0301–PP_0305 operon, and thereby channels carnitine-derived carbon and nitrogen into the conserved glycine-betaine demethylation route. The genome-legacy name "betainyl-CoA thiolase" (EC 2.3.1.-) is a misannotation; the enzyme is a thioester hydrolase, not a thiolase.